

LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data **Student:** Akshaya Gadeela **Date:** 05 October 2026

Objective

Predict **Outcome:** 0 = No diabetes, 1 = Diabetes. The project demonstrates a complete binary-classification workflow.

Dataset & Workflow

Pima Indians Diabetes Database (supplied CSV). Inputs: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction and Age. Workflow: load data → inspect data → replace invalid zeros with missing values → train-test split → median imputation → StandardScaler → balanced Logistic Regression → evaluate → interpret coefficients.

Model

Median imputation + StandardScaler + Logistic Regression with **class_weight='balanced'**, 80:20 stratified split, fixed random_state = 111.

Evaluation Metrics

Metric	Result	Meaning
Accuracy	0.734	Overall correct predictions
Precision	0.603	Reliability of positive predictions
Recall	0.704	Actual positive cases detected
F1-score	0.650	Balance of precision and recall
ROC-AUC	0.831	Overall class-separation ability

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 75	FP = 25
Actual 1	FN = 16	TP = 38

Key Findings

The model detects approximately 70.4% of positive cases in the fixed test split. Recall is especially important in a screening-style classification task because false negatives represent missed positive cases. The standardized coefficient plot provides an interpretable view of feature contributions.

Conclusion

I built and evaluated a Logistic Regression model for diabetes prediction using a reproducible preprocessing and classification pipeline. The project improved my understanding of data preparation, model training, confusion matrices, ROC-AUC and coefficient interpretation. This model is intended for learning classification concepts, not medical diagnosis.

Submission: Keep the executed .ipynb outputs and graphs visible. Add any required student ID/batch details before uploading.